Paper 00 · Starting from Two Facts: An Introduction to the Digital Life Series

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Digital Life Series · Paper 0

Date: 2026-10-23

Note: This paper is the introduction to the series: it lays out two facts, gives one definition, and explains the structure of the series. The full derivation is in Paper 1.

English translation of the Chinese original. The Chinese text is authoritative: https://doi.org/10.5281/zenodo.23591398

English compendium: https://doi.org/10.5281/zenodo.23592082

Series page: https://github.com/heqiq619/ai-not-alive

Abstract

This paper does not argue; it does only four things: lay out two facts, give one definition, set out the special case of the digital world, and then use it to look at today's AI. The two facts are: the p-language of language can be approximated by functions; the laws of physics can be described mathematically. The definition is: write the universe as two parts (universe = P + R), and write life as two parts as well (life = p + R-life); p is a self-referential structure that lies in P and is preserved (maintainable), and R is the non-functionizable part. This paper is the introduction; the checklist, the three necessary conditions, and the discussion of emergence are all in Paper 1.

Notation

- **P**: the functionizable part of the universe.

- **R-universe**: the non-functionizable part of the universe; this paper only declares that it may exist, and does not develop it.

- **p-life**: the part of life that falls in P — the object this paper defines and discusses.

- **R-life**: the part of life that falls in R; only declared, not developed.

- **p-environment / R-environment**: the two parts of the environment, defined in the same way.

- **p-language / R-language**: the two parts of language; what large models compress is p-language.

- **Residual**: the form in which R's influence on the p level manifests — a lower bound on error that cannot be eliminated at the p level.

From here on we use shorthand throughout: when we say "the universe" we mean P, when we say "life" we mean p-life, and when we say "the environment" we mean p-environment.

1. Two Facts

The first fact: the laws of physics can be described mathematically. The evidence is existence — physics itself is this.

The universe can be written as a decomposition:

> universe = P + R-universe

P is the functionizable part; R-universe is the non-functionizable part. The part described by the laws of physics falls in P. So at least we can say: P is non-empty.

The second fact: part of language can be approximated by functions. The evidence is not reasoning but existence — large models, using finite parameters and a computable process, generate text that is statistically close to human language.

Language can also be written as a decomposition:

> language = p-language + R-language

p-language is the functionizable part of language, and it is what large models compress; R-language is the non-functionizable part of language, which function approximation cannot reach. So the accurate statement is not "language can be approximated by functions", but "the p-language of language can be approximated by functions"; about R-language we can only remain silent.

Together the two facts show: the universe is not entirely non-functionizable. There exists at least one non-empty functionizable part, denoted P. This layer is not a conjecture but an inference from facts; it says only that P exists, not how large P is.

The boundary between P and R is absolute: an object is either functionizable (belongs to P) or non-functionizable (belongs to R); there is no in-between; $P \cap R = \varnothing$, $P \cup R$ = the universe.

If R has an influence on the p level, it will not be described directly by the language of functions; it will show up only as a residual that cannot be eliminated at the p level: no matter how much more data and capacity are added, the error stays at some positive number. This lower bound on error is the projection of R onto the p level.

For convenience in what follows, we abbreviate it as:

> universe = P + residual

> life = p-life + residual

> environment = p-environment + residual

What follows discusses only the P side; the residual is mentioned only when needed, and its content is not developed.

2. One Definition

First write the universe as two parts:

> universe = P + R-universe

P is the functionizable part, and R-universe is the non-functionizable part of the universe. The two facts guarantee that P is non-empty; how large R-universe is, this paper does not know and does not claim.

On this decomposition, life itself is also written as two parts:

> life = p + R-life

Here p is the functionizable part: a self-referential structure that lies in the universe's P and is preserved (maintainable); R-life is the non-functionizable part of life itself: this paper does not know what it is, and does not claim that it does not exist. Note that the two R's are not the same thing — R-universe is the non-functionizable part at the level of the universe, while R-life is the non-functionizable part inside a single life.

Three things define p-life. First, scope: p is sought only in the universe's P (by the same token, language also divides into p-language and R-language, and what large models approximate is p-language).

Second, preserved — that is, maintained. Here we must make clear a place that is easy to get backwards: maintenance is not an attribute of life itself but a relation between life and the universe. Whether a structure can be preserved depends on whether its orbit stays inside the universe's maintainable set; this condition is written in the universe, and is not announced by the structure itself. So "preserved" is the name of a relation, not a third coordinate attribute standing alongside the others.

Third, self-reference: the update rule is inside the system itself. This is an attribute internal to life, and it is also where it differs from an ordinary program.

What about R-life? It can only be handled as an approximation, with a lower bound on error attached. So the criteria and measurements of this paper cover p-life, not the whole of life; about R-life, this paper says only "the approximation stops here".

The definition is made, not discovered; whether it holds depends on whether it can distinguish, and whether it can derive something measurable.

One point needs to be stated: as for the R part, this series only declares its existence here — the universe, life, the environment, and language can all be written as p + R, but what R is and how large it is, this paper does not know and does not claim. For ease of understanding, what follows by default discusses only the p side: whenever we say "the universe", we mean P; whenever we say "life", we mean p-life; whenever we say "the environment", we mean p-environment. What follows uses "residual" to denote the influence of R; the concrete content of R is no longer developed.

3. The Digital World: Fully Functionizable

The digital world is fully functionized: it consists of bits, states, and transition rules, and has no part that a function description cannot reach. As a formal object, its absolute R = ∅ — this is guaranteed by construction: every property corresponds to a mathematical function, even if that function is not computable.

Digital life is part of the digital world, so its R-life can only be the empty set as well: digital life = p-life, fully functionizable.

Whether absolute R exists, this paper does not judge (see the supplementary section). For digital life, there is truly only one decisive question: **must life have an absolute R?** There are only two outcomes.

The first outcome, life can lack an absolute R: then digital life is possible — it consists entirely of p-life and needs no absolute R at all.

The second outcome, life must have an absolute R: then digital life is impossible — the digital world cannot produce an absolute R.

This paper does not judge which outcome holds. What can be certain is: whichever it is, this paper's definition and measurement of p hold; the only difference is whether digital life is possible or impossible.

One more sentence, on something easy to confuse: the above concerns absolute R; for any finite observer, relative R is non-empty (see the supplementary section), and it does not affect this judgment. In addition, functionizable

does not equal measurable — measurement also requires computability, observability, and reproducibility. The digital world has all of these, so measurement is feasible in the digital world; for the universe as a whole, even if it can be described by functions, we cannot measure it, because we live inside it.

4. Using It to Look at Today's AI

Under the present definition, AI lacks two things. Its update rule is on the outside, so it is not self-referential; its maintenance is also not its own affair — it is maintained by external power supply and operations, which is its relation to the universe, not its own capacity.

So it is a very strong decompressor, but once it has been written it just stops there — a frozen decompressor. It has no R-life: in the digital world $R = \varnothing$, so AI is a pure-p decompressor.

5. The Place of This Introduction

This paper gives only facts, definitions, and conclusions. The derivation details, the checklist, the three necessary conditions, and the discussion of emergence are all in Paper 1; the measurement is in Paper 2; the experiments are in Papers 4 through 9; the constraints and safeguards are in Papers 10 and 11. The whole series is arranged in this order.

> This paper was written by He Qi, with DeepSeek (AI) assisting in the arrangement; the views and the responsibility belong to the author.

Supplement (Optional Reading): The Domain and Codomain of P

The main line ends with the previous section; this section adds only one formal detail: what exactly the "domain" and "codomain" of P are. Not reading it does not affect the main line.

The universe (the P side) can be regarded as a state space S: all states that can be described by functions. The universe function is its mapping to itself, written $F: S \to S$. So the domain of P is S, and the codomain is S as well — it maps to itself.

An observer O is just a subsystem inside S: the input it can read is a subspace of S, denoted D_O; the output it can write is another subspace, denoted C_O. Both D_O and C_O are contained in S. The observer's "functionizability" (P_O) is just this: between D_O and C_O, there exists a family of functions whose error can approach zero.

R has no domain or codomain to speak of: domain/codomain are concepts in the language of functions, and R is the non-functionizable side. So of R one cannot ask "what is its domain" — this question itself uses the wrong language.

Where what follows involves "domain/codomain", it means by default the observer-side D_O and C_O; for P at the universe level, it means by default the self-mapping S → S.

Why Relative R Is Non-Empty (Optional Reading)

If the domain/codomain of the world fluctuates within the observer's update period (fast variables), the observer cannot keep up with these changes; that part is non-functionizable for it, and so relative R is non-empty. This is a sufficient condition: there are fast-variable fluctuations ⇒ relative R is non-empty. We have not said that it must hold, and we do not explain the causes of the fluctuations.

Two further strengthening reasons (either one suffices): first, finite resources — cannot see everything, cannot compute everything; second, self-reference — cannot fully predict itself (a system that can always predict its own next step can do the opposite, and thereby contradict itself).

Whether absolute R exists, this paper does not judge.

The same holds for the digital world: inside it, too, there must be relative R. The same sufficient condition — short-term fluctuations of the domain/codomain; and in addition three further, mutually independent sufficient conditions: undecidability, computational irreducibility, self-reference (cannot fully predict itself). Here two sentences must be distinguished: "the digital world R = ∅" is a statement at the ontological level; "relative R is non-empty" is a statement at the observer level — fully functionizable does not equal fully knowable.

One last sentence about the boundary: if the essence of life lies on the absolute R side, this paper — and any text written in the language of functions — cannot give its definition. This is not an oversight but the boundary of the language of functions: the definition itself is on the P side, and to define R in the language of P is to pull it back into P.